

L -functions for all meromorphic $\mathrm{SL}_2(\mathbb{Z})$ modular forms, via manifestly finite contours

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Abstract

We give a single finite, regulator-free functional realizing in one object the period polynomials (at integer s) and the completed L -functions (at general s) of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$, with poles anywhere: at the cusp and in the interior. It is a two-segment contour integral with an explicit Hurwitz-zeta T -kernel, built from any finite base point $\tau_0 \in \mathbb{H}$ and manifestly finite. For weakly holomorphic forms it is independent of τ_0 and of the path, reproducing at $\tau_0 = i$ the special L -values of Bringmann–Fricke–Kent [1]; its integer- s values give the period polynomial, Brown’s quasi-periods [2] in closed form. For forms with interior poles the value depends only on the homotopy class of the contour — fixed up to the residues it crosses — and a natural choice of class, the one prescription a pole on the line $\mathrm{im} \tau = 1$ requires, renders it canonical. The resulting closed form generalizes [3]: a regularized cusp sum plus one polylog integral per interior pole. The construction introduces no new objects, unifying these periods and L -values through a single all- s kernel, and every formula is checked numerically [4].

1 A finite L -integral

Definition 1.1 (S - and T -segments). *Let $\tau_0 \in \mathbb{H}$ and $f \in F_k$, let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ denote the generators of $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, and let $f^{-1}(\infty)$ be the locus of all poles of $f \in \mathbb{H}$ (inclusive of the cusp of $\mathrm{SL}_2(\mathbb{Z})$). The S - and T -segments at τ_0 are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$,*

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0, \quad (1)$$

running from $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 .

Note: When $f \in M_k^!$, avoiding poles is automatic, as long as τ_0 , i.e. the base-point of any of these paths is not at a cusp of $\mathrm{SL}_2(\mathbb{Z})$, i.e. there is only the trivial homotopy class. However, when $f \in F_k$, and f has poles at finite and non-zero imaginary heights, then these integration contours fall into nontrivial homotopy classes.

Definition 1.2 (Branch prescription for exponentials). *Let e^x denote the exponential map from $\mathbb{C} \rightarrow \mathbb{C}$. It is helpful to define the following shorthand: $e(x) := e^{2\pi i x}$*

Definition 1.3 (Hurwitz T -kernel). *Let $\zeta(s, x) = \sum_{m \geq 0} (m + x)^{-s}$ as a function $\zeta : \mathbb{C}_s \times (\mathbb{C} \setminus \{0\})_x \rightarrow \mathbb{C}$ denote the Hurwitz zeta function (meromorphic in s , holomorphic in $x \in \mathbb{H}$). We define Hurwitz T -kernel as*

$$\tilde{k}_T(\tau, s) := \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(1 - (k - s), \tau + 1). \quad (2)$$

Definition 1.4 (L -integral). *Let $f \in F_k$, $\tau_0 \in \mathbb{H}$, and $s \in \mathbb{C}$. Further, let $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ be the S - and T -segments at τ_0 , as in Definition 1.1, and let $e(x)$ and $\tilde{k}_T$ be as defined in Definitions 1.2 and 1.3. The L -integral of f is*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := e^{-\pi i s/2} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right]. \quad (3)$$

Path-dependence and τ_0 -independence of $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ are dealt with below, in the main body. But, first, we use this L -integral to define a generalized L -function:

Definition 1.5 (L -function). *Let $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ be as defined in 1.4. We define the generalized L -function of $f \in F_k$ to be,*

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (4)$$

In this vein, τ_0 -independence and path-dependence of the L -integral and the L -function of $f \in F_k$ are in lockstep. For $f \in F_k$ such that $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is both τ_0 -independent and path-independent, then we may write $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L^*(f, s)$ and similarly for $L(f, s)$. First, though, finiteness:

Lemma 1.6 (Finiteness). *For $f \in F_k$, both integrals in Definition 1.4 are finite for every $s \in \mathbb{C}$, every $\tau_0 \in \mathbb{H}$, and every choice of paths $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$.*

Proof. Both segments are compact and lie in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, so f is continuous on them; τ^{s-1} is continuous on $\mathbb{H}$, and $\tilde{k}_T(\tau, s)$ is continuous there since $\zeta(\nu, \tau + 1)$ is holomorphic in $\tau \in \mathbb{H}$ (Definition 1.3). A continuous integrand on a compact path is integrable, for every $s \in \mathbb{C}$. $\square$

Proposition 1.7 (Basepoint and path independence). *For $f \in M_k^!$ the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the paths: with no interior poles, $\mathbb{H}$ is simply connected, so all admissible contours are homotopic. We write $L^*(f, s)$ for the common value and $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$.*

Proof. With no interior poles the integrands $f(\tau)\tau^{s-1}$ and $f(\tau)\tilde{k}_T(\tau, s)$ are holomorphic on $\mathbb{H}$, which is simply connected; by Cauchy's theorem each integral is unchanged under any deformation fixing its endpoints, hence depends on those endpoints alone. This is path-independence.

For independence of τ_0 we differentiate the two integrals and watch them cancel. The T -segment runs from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 — two endpoints a unit apart, moving together with τ_0 — so by Leibniz's rule and the T -periodicity $f(\tau_0 - 1) = f(\tau_0)$,

$$\partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) \tilde{k}_T(\tau_0, s) - f(\tau_0 - 1) \tilde{k}_T(\tau_0 - 1, s) = f(\tau_0) \delta_T \tilde{k}_T(\tau_0 - 1, s), \quad (5)$$

where $\delta_T \tilde{k}_T(\tau, s) := \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s)$. This is the whole role of δ_T : the unit gap between the endpoints turns the ∂_{τ_0} boundary term into a unit shift of the kernel. That shift is elementary — the kernel was built for it — since $\zeta(\nu, x + 1) - \zeta(\nu, x) = -x^{-\nu}$ gives, on both terms at $x = \tau + 1$,

$$\delta_T \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s}, \quad \text{hence} \quad \partial_{\tau_0} \int_{\gamma^T} f \tilde{k}_T = f(\tau_0) [-\tau_0^{s-1} + e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (6)$$

The S -segment runs $-1/\tau_0 \rightarrow \tau_0$; its lower endpoint moves with $\partial_{\tau_0}(-1/\tau_0) = \tau_0^{-2}$, so by Leibniz, the modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, and the phase $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)}\tau_0^{1-s}$,

$$\partial_{\tau_0} \int_{\gamma^S} f \tau^{s-1} = f(\tau_0) \tau_0^{s-1} - f(-1/\tau_0) (-1/\tau_0)^{s-1} \tau_0^{-2} = f(\tau_0) [\tau_0^{s-1} - e^{i\pi(s-1)}\tau_0^{k-1-s}]. \quad (7)$$

The two brackets are negatives of each other, so $\partial_{\tau_0} L^* = e^{-\pi i s/2} (\partial_{\tau_0} \int_{\gamma^S} + \partial_{\tau_0} \int_{\gamma^T}) = 0$. $\square$

Remark 1.8 (Classical Mellin recovery). *For a holomorphic cusp form $f \in S_k$, the cusp limit $\tau_0 \rightarrow i\infty$ collapses the S -segment onto $[0, i\infty]$ while the T -segment vanishes exponentially, so*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt, \quad (8)$$

the classical completed L -integral.

2 The L -function

Proposition 2.1 (Hurwitz-Gamma). *For $n \neq 0$ and $\nu \in \mathbb{C}$,*

$$\int_i^{i+1} e^{2\pi i n u} \zeta(\nu, u) du = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (9)$$

Proof. Write $\zeta(\nu, u) = \sum_{m \geq 0} (m+u)^{-\nu}$ and shift $u \mapsto u+m$ in the m -th term; as $e^{2\pi i n(u+m)} = e^{2\pi i n u}$ for integer m , the unit segments stack into one ray,

$$\int_i^{i+1} e^{2\pi i n u} \zeta(\nu, u) du = \sum_{m \geq 0} \int_{i+m}^{i+1+m} e^{2\pi i n v} v^{-\nu} dv = \int_i^{i+\infty} e^{2\pi i n v} v^{-\nu} dv. \quad (10)$$

For $n > 0$ the integrand is holomorphic in the first quadrant and decays as $\text{im } v \rightarrow \infty$ (since $|e^{2\pi i n v}| = e^{-2\pi n \text{im } v}$), so the horizontal ray rotates to the vertical $[i, i\infty)$. With $v = it$ ($t \geq 1$; then $dv = i dt$, $e^{2\pi i n v} = e^{-2\pi n t}$, $v^{-\nu} = e^{-\pi i \nu/2} t^{-\nu}$) and $w = 2\pi n t$,

$$\int_i^{i+\infty} e^{2\pi i n v} v^{-\nu} dv = e^{i\pi(1-\nu)/2} \int_1^\infty e^{-2\pi n t} t^{-\nu} dt = e^{i\pi(1-\nu)/2} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (11)$$

For $n < 0$ the integrand instead grows as $\text{im } v \rightarrow \infty$, so the closing arc survives and the rotation is illegal; but both sides are analytic in n and agree for $n > 0$, so the identity continues to $n < 0$, with $\Gamma(1-\nu, 2\pi n)$ on its principal branch. At $n = 0$ the prefactor $(2\pi n)^{\nu-1}$ is singular — that mode, the constant Fourier coefficient, is handled separately. $\square$

Theorem 2.2 (L -function for weakly holomorphic forms). *For $f \in M_k^!$, at $\tau_0 = i$,*

$$L^*(f, s) = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right] = i^k L^*(f, k-s). \quad (12)$$

Proof. At $\tau_0 = i$ the S -segment degenerates ($-1/i = i$), so $L^*(f, s) = e^{-\pi i s/2} \int_{i-1}^i f \tilde{k}_T d\tau$. Expand $f = \sum_n c_f(n) e^{2\pi i n \tau}$ and substitute $u = \tau + 1$ ($e^{2\pi i n(u-1)} = e^{2\pi i n u}$): each $n \neq 0$ mode is handled by Proposition 2.1. The first kernel term with $\nu = 1-s \leftrightarrow 1-\nu = s$ gives $e^{-\pi i s/2} e^{\pi i s/2} (2\pi n)^{-s} \Gamma(s, 2\pi n) = \Gamma(s, 2\pi n)/(2\pi n)^s$; the second kernel term with $\nu = s-k+1 \leftrightarrow 1-\nu = k-s$ gives

$$-e^{-\pi i s/2} e^{i\pi(s-1)} e^{\pi i(k-s)/2} (2\pi n)^{s-k} \Gamma(k-s, 2\pi n) = i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}, \quad (13)$$

the s cancelling in the phase $-e^{-\pi i s/2} e^{i\pi(s-1)} e^{\pi i(k-s)/2} = e^{\pi i k/2} = i^k$. So each $n \neq 0$ mode contributes $\Gamma(s, 2\pi n)/(2\pi n)^s + i^k \Gamma(k-s, 2\pi n)/(2\pi n)^{k-s}$.

Crucially, the $n = 0$ mode contributes $c_f(0) e^{-\pi i s/2} \int_{i-1}^i \tilde{k}_T d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right)$.

Swapping the two Γ -terms and using $i^{2k} = (-1)^k = 1$ (the weight k is even) shows that bracket equals i^k times its $s \mapsto k-s$ image; the polar term is i^k -symmetric likewise, giving $L^*(f, s) = i^k L^*(f, k-s)$. $\square$

2.1 Period polynomials, periods and quasi-periods

Definition 2.3 (Period cocycle). *Let $V_n := \mathbb{C}[X, Y]_n$ with the action $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$, and $\omega_f := f(\tau) (X - \tau Y)^n d\tau$. For $f \in F_k$ and $\tau_0 \in \mathbb{H}$, the period cocycle is*

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f \in V_n, \quad \gamma \in \Gamma. \quad (14)$$

From this definition, it is straightforward to see that $C_{gh}^f = C_g^f|_h + C_h^f$. The proof is straightforward and canonical: Split the contour of C_{gh}^f at $h^{-1}\tau_0$: $\int_{h^{-1}g^{-1}\tau_0}^{\tau_0} \omega_f = \int_{h^{-1}g^{-1}\tau_0}^{h^{-1}\tau_0} \omega_f + \int_{h^{-1}\tau_0}^{\tau_0} \omega_f$. The second term is C_h^f ; by the Γ -equivariance of ω_f ($f|_k h = f$ gives $\int_{h^{-1}a}^{h^{-1}b} \omega_f = (\int_a^b \omega_f)|_h$) the first is $C_g^f|_h$, whence $C_{gh}^f = C_g^f|_h + C_h^f$.

Lemma 2.4 (Parabolic representatives exist iff $c_f(0) = 0$). *With $\delta_T P := P|_T - P$ on V_n , one has $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0)$, so $C_T^f \in \text{im } \delta_T$ if and only if $c_f(0) = 0$ — the principal part of f never obstructs.*

Proof. The X^n -coefficient of ω_f is $f(\tau) d\tau$, so $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} f d\tau = (2\pi i)^{n+1} c_f(0)$, the $m \neq 0$ Fourier modes integrating to zero over a unit period. On the other side, $(P|_T)(X, Y) = P(X+Y, Y)$ has the same X^n -coefficient as P , so $[\delta_T P]_{X^n} = 0$ for every $P \in V_n$; since $\ker \delta_T = \mathbb{C} Y^n$ is one-dimensional, $\text{im } \delta_T$ is the entire hyperplane $\{Q : [Q]_{X^n} = 0\}$. Hence $C_T^f \in \text{im } \delta_T$ iff $[C_T^f]_{X^n} = 0$ iff $c_f(0) = 0$. $\square$

Definition 2.5 (Period polynomial). *For $f \in S_k^!$ (so $c_f(0) = 0$), the period polynomial is the S -value of the parabolic representative,*

$$r_f := C_S^f(\tau_0) - (P|_S - P) \in V_n, \quad (15)$$

with $P \in V_n$ chosen so that $\delta_T P = C_T^f$ (Lemma 2.4).

Proposition 2.6 (Period coefficients are special L -values). *For $f \in S_k^!$ and $\ell \in \{1, \dots, k-3\}$,*

$$[r_f]_{X^{n-\ell} Y^\ell} = (2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1). \quad (16)$$

Proof. The cocycle is the coboundary of the Eichler primitive $\tilde{F}(\tau) = \int^\tau f(z)(X - zY)^n dz$, namely $C_\gamma^f = \tilde{F} - \tilde{F}|_\gamma$, and $r_f = C_S^f - (P|_S - P)$ is the S -value of its cusp-regular (T -flat) branch, P solving $\delta_T P = C_T^f$ (Lemma 2.4). Read the $X^{n-\ell} Y^\ell$ -coefficient. From $(X - \tau Y)^n = \sum_j \binom{n}{j} (-\tau)^j X^{n-j} Y^j$,

$$[C_S^f]_{X^{n-\ell} Y^\ell} = (2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} \int_{\gamma_S} f \tau^\ell d\tau. \quad (17)$$

The kernel realizes the correction: $\tilde{k}_T(\tau, \ell+1)$ is the antidifference $\delta_T \tilde{k}_T(\tau, \ell+1) = -(\tau+1)^\ell + (-1)^\ell (\tau+1)^{n-\ell}$, whose two monomials are the ℓ - and $(n-\ell)$ -components exchanged by $Y|_S = X$; so the T -segment integral against $\tilde{k}_T$ is the coboundary P inverting δ_T , with $[P|_S - P]_{X^{n-\ell} Y^\ell} = -(2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} \int_{\gamma_T} f \tilde{k}_T(\tau, \ell+1)$. Therefore

$$[r_f]_{X^{n-\ell} Y^\ell} = (2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} \left[\int_{\gamma_S} f \tau^\ell + \int_{\gamma_T} f \tilde{k}_T(\tau, \ell+1) \right] = (2\pi i)^{n+1} (-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1), \quad (18)$$

using $L^*(f, \ell+1) = i^{-(\ell+1)} [\int_{\gamma_S} f \tau^\ell + \int_{\gamma_T} f \tilde{k}_T(\tau, \ell+1)]$. $\square$

Quasi-periods and their determinant: As an aside, the Hurwitz T -kernel in this note was originally uncovered in order to reproduce the manifestly finite computations of the quasi-periods for $\hat{\Delta} = 1/q + \mathcal{O}(q^2) \in S_{12}^!$ originally given in [2]. In this vein, it is interesting to show collect quasi-periods computable from our L -integrals/period-polynomials for $f \in S_k^!$ (when $\dim(S_k) = 1$):

Consider the case of a weak cusp form $f \in S_k^!$ for k such that $\dim S_k = 1$, i.e. $k \in \{12, 16, 18, 20, 22, 26\}$. The coefficients of r_f (Proposition 2.6) are Brown's quasi-period coordinates $\omega^\pm(f)$ [2]: ω^+ from ℓ even, ω^- from ℓ odd. Taking $f = \widehat{\Delta}_k := q^{-1} + O(q^3) \in S_k^!$, our period polynomials reproduce Brown's published $\eta^\pm(\widehat{\Delta}_{12})$ at $k = 12$ and give related quasi-periods at the remaining five weights (Table 1). Their determinant satisfies a Legendre-type relation, $\omega^+\eta^- - \omega^-\eta^+ \in \mathbb{Q}^\times \cdot (2\pi i)^{k-1}$, equal to $-(k-2)!(2\pi i)^{k-1}$ at $k = 12$ in Brown's normalization (the weight- k analog of $\omega_1\eta_2 - \omega_2\eta_1 = 2\pi i$). Both are global consistency checks, claiming nothing new [4]. *this paragraph needs to be drastically de-LLM-ized!!!*

k	$\omega^+(f_k)$	$\omega^-(f_k)/i$	$\eta^+(\widehat{\Delta}_k)$	$\eta^-(\widehat{\Delta}_k)/i$	$\det / (2\pi i)^{k-1}$
12	-6.89168e7	-5.58502e6	1.27202e11	1.02767e10	-3628800
16	-8.81481e11	-1.54063e10	4.14120e13	6.83541e11	-37777259520
18	-5.07833e13	-5.43305e12	-2.19200e14	-3.36339e13	13948526592000
20	-4.22882e16	-3.73170e14	6.38640e16	-1.89823e14	-21768070599475200
22	-1.49974e18	-4.24489e16	1.43795e18	-4.41050e16	2201197055016960000
26	-1.18673e22	-3.60961e21	1.18653e22	-3.61016e21	951354215990967140352000

Table 1: Periods $\omega^\pm(\Delta_k)$ of $\Delta_k = E_{k-12}\Delta$ (and $E_0 = 1$) and quasi-periods $\eta^\pm(\widehat{\Delta}_k)$ of the weak cusp form $\widehat{\Delta}_k = \Delta_k(J_2 + a_{k,1}J_1 + a_{k,0}) = 1/q + O(q^2) \in S_k^!$ (where $a_{k,1}, a_{k,0}$ are rational numbers that ensure the q -series gap and $J_k = 1/q^k + O(q) \in S_0^!$), and the integer $\det / (2\pi i)^{k-1}$ with $\det = \omega^+\eta^- - \omega^-\eta^+$. Full 50-digit values in [4]. *this paragraph needs to be drastically de-LLM-ized!!!*

2.2 Why a finite contour

For a weak cusp form $f \in S_k^!$ the classical Mellin integral $\int_0^{i\infty} f(\tau)(\tau/i)^s \frac{d\tau}{\tau}$ does not converge — f grows at the cusp — and the deformed-Mellin construction of [1] supplies a finite value through a non-holomorphic regularization there. The functional of Definition 1.4 reaches the same value with no regularization: written as its two segments, each diverges as $\tau_0 \rightarrow i\infty$, with opposite signs, and the two cancel at every finite τ_0 , so $L^*(f, s)$ is finite throughout $\mathbb{H}$ although the Mellin contour itself has no limit. By Proposition 1.7 the value is independent of τ_0 and of the paths; BFK's regularized value is its evaluation at $\tau_0 = i$ (Theorem 2.2), and its integer- s values are Brown's quasi-periods. Two independent finite computations — Brown's cohomological periods and the BFK L -values, obtained without reference to one another — meet at this one contour, which fixes it as the canonical object. Finiteness and τ_0 -independence are the analytic shadow of the parabolic representative [2, 6, 7]. *this paragraph needs to be drastically de-LLM-ized!!!*

3 Meromorphic modular forms

Proposition 3.1 (Basepoint and path independence within a homotopy class). *For $f \in F_k$, the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of τ_0 and of the path so long as both vary within a single homotopy class of admissible contours in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$. For $f \in M_k^!$ there is one class and Proposition 1.7 is recovered; the dependence between classes is the residue crossing of Corollary 3.10.*

Proof. The τ_0 -derivative of Proposition 1.7 uses only the T -periodicity $f(\tau_0 - 1) = f(\tau_0)$ and the modularity $f(-1/\tau_0) = \tau_0^k f(\tau_0)$, both valid for every $f \in F_k$, and is evaluated at the moving endpoints alone. While τ_0 stays in one chamber of $\mathbb{H} \setminus f^{-1}(\infty)$ — where the segments meet no pole — the boundary terms cancel exactly as there, so $\partial_{\tau_0} L^* = 0$; the interior poles lie off the contour

and never enter. Within a fixed class the integrands $f\tau^{s-1}$ and $f\tilde{k}_T$ are holomorphic along the contours, so any two are joined by a deformation avoiding $\Gamma\cdot f^{-1}(\infty)$, and Cauchy's theorem equates their integrals. $\square$

Definition 3.2 (Polylog projection). *Let $f \in F_k$ have interior poles at the points τ_p of a set $\mathcal{P}_f$ of Γ -inequivalent representatives, and write $\text{Li}_s(z) = \sum_{n>0} z^n n^{-s}$. Since $\text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$ has a pole of order m at τ_p , the principal part of f there is matched by a unique polylog projection*

$$\hat{P}_{\tau_p} f(\tau) := \sum_{m>0} r_{f,\tau_p}^*(m) \text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)}), \quad (19)$$

its coefficients $r_{f,\tau_p}^(m)$ being the principal-part data of f at τ_p . The projection $\hat{P} := \sum_{\text{Im } \tau_p \geq 1} \hat{P}_{\tau_p}$ removes the principal part of every pole on or above the line $\text{Im } \tau = 1$.*

Definition 3.3 (Regularized cusp coefficients). *With $\hat{P}$ as in Definition 3.2,*

$$c_{Rf}(n) := [(I - \hat{P})f]_{q^n}. \quad (20)$$

Definition 3.4 (The space $F(\infty)$). *$F(\infty)$ is the space of meromorphic g on $\mathbb{H}$ with $g(\tau) = g(\tau+1)$, finitely many poles in any compact subset of the strip $\{\text{Im } \tau > 0, |\text{Re } \tau| \leq \frac{1}{2}\}$, and $g(\tau) = O(e^{C \text{Im } \tau})$ as $\text{Im } \tau \rightarrow \infty$ for some C . Each such g has a Fourier expansion $g = \sum_n c_g(n)q^n$.*

Then $F_k \subset F(\infty)$, and $(I - \hat{P})f \in F(\infty)$ — the polylog tails are T -periodic but not modular, so the projection leaves F_k .

Lemma 3.5 (q -series–integral commutation). *Let $g \in F(\infty)$ have no pole on or above the line $\text{Im } \tau = 1$ (e.g. $g = (I - \hat{P})f$). Then $g = \sum_n c_g(n)q^n$ converges along the T -segment at $\tau_0 = i$ and integrates term by term:*

$$\int_{\gamma_i^T} g(\tau) \tilde{k}_T(\tau, s) d\tau = \sum_n c_g(n) \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau. \quad (21)$$

A pole at $\text{Im } \tau_p \geq 1$ defeats this — there the coefficients grow at least like $n^{m-1}e^{2\pi n \text{Im } \tau_p}$ [3, Lem. 2.6].

Proof. With no pole on or above $\text{Im } \tau = 1$, the finitely many poles of g in the strip have heights < 1 ; let $h < 1$ be the largest. The Fourier series $g = \sum_n c_g(n)q^n$ converges absolutely and uniformly on compact subsets of $\{\text{Im } \tau > h\}$, in particular on the T -segment $[i-1, i]$; with $\tilde{k}_T(\tau, s)$ bounded there, the sum passes through the integral, which is the displayed identity. Conversely a pole at height $\text{Im } \tau_p \geq 1$ forces $|c_g(n)| \gtrsim n^{m-1}e^{2\pi n \text{Im } \tau_p}$ [3, Lem. 2.6], so $|c_g(n)q^n| \gtrsim n^{m-1}e^{2\pi n(\text{Im } \tau_p - 1)} \not\rightarrow 0$ on $[i-1, i]$ and the series diverges. $\square$

Definition 3.6 (Polylog integral). *For $N \geq 0$ and $z \in \mathbb{H}$, the polylog integral is the L -integral of a single tail $\text{Li}_{-N}(e^{2\pi i(\tau-z)})$ along the T -segment at $\tau_0 = i$ (where γ^S degenerates, since $-1/i = i$):*

$$I_N(s, z) := e^{-\pi i s/2} \int_{i-1}^i \text{Li}_{-N}(e^{2\pi i(\tau-z)}) \tilde{k}_T(\tau, s) d\tau. \quad (22)$$

Lemma 3.7 (Closed form of the polylog integral). *Write $\mathcal{G}(s, m) := \frac{\Gamma(s, 2\pi m)}{(2\pi m)^s} + i^k \frac{\Gamma(k-s, 2\pi m)}{(2\pi m)^{k-s}}$, the building block of Theorem 2.2 continued to $m < 0$. For the projected poles $\text{Im } z > 1$, where*

$|e(\tau - z)| \geq 1$ on the contour, polylog inversion $\text{Li}_{-N}(w) = (-1)^{N+1} \text{Li}_{-N}(1/w)$ resums the integral into the convergent series

$$I_N(s, z) = \delta_{N,0} \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + (-1)^{N+1} \sum_{n \geq 1} n^N e^{2\pi i n z} \mathcal{G}(s, -n). \quad (23)$$

Proof. For $\text{Im } z > 1$, $|e(\tau - z)| \geq 1$ on $[i-1, i]$, so the series of $\text{Li}_{-N}(e(\tau - z))$ diverges; the inversion rewrites it through $\text{Li}_{-N}(e(z - \tau)) = \sum_{n \geq 1} n^N e^{2\pi i n(z - \tau)}$, convergent since $|e(z - \tau)| \leq 1$. Integrating term by term against $\tilde{k}_T$, each $e^{-\pi i s/2} \int_{i-1}^i e^{-2\pi i n \tau} \tilde{k}_T d\tau$ is the building block $\mathcal{G}(s, -n)$ of Theorem 2.2 at mode $-n$, giving the displayed sum; for $N = 0$ the leftover -1 in $\text{Li}_0(w) = -1 - \text{Li}_0(1/w)$ adds $-e^{-\pi i s/2} \int_{i-1}^i \tilde{k}_T = \frac{1}{s} + \frac{i^k}{k-s}$. $\square$

Theorem 3.8 (*L*-function of a meromorphic modular form). *For $f \in F_k$ with no pole on the edge $\text{Im } \tau = 1$, at $\tau_0 = i$,*

$$L^*(f, s) = -c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) \mathcal{G}(s, n) + \sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p), \quad (24)$$

$\mathcal{G}(s, n) = \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}}$; the regularized cusp sum (Definition 3.3, Theorem 2.2) plus one polylog integral (Lemma 3.7) per interior pole above the edge. Poles on the edge enter by the principal value of Lemma 3.9.

Proof. At $\tau_0 = i$ the S -segment degenerates, so $L^*(f, s) = e^{-\pi i s/2} \int_{i-1}^i f \tilde{k}_T d\tau$. Split $f = (I - \hat{P})f + \hat{P}f$. The regularized part has no pole on or above $\text{Im } \tau = 1$ (Definition 3.2), so by Lemma 3.5 its Fourier series integrates term by term; mode by mode this is Theorem 2.2, giving $-c_{Rf}(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) \mathcal{G}(s, n)$. The projected part $\hat{P}f = \sum_{\tau_p} \sum_{m > 0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e(\tau - \tau_p))$ integrates to $\sum_{\text{Im } \tau_p > 1} \sum_{m > 0} r_{f, \tau_p}^*(m) I_{m-1}(s, \tau_p)$ by Lemma 3.7. Their sum is the stated formula. $\square$

Lemma 3.9 (*L*-function, a pole on the edge). *Routing the T -path just below versus just above a pole τ_p of $f \in F_k$ at height $\text{Im } \tau_p = 1$ gives finite values differing by the residue,*

$$L_{\text{below}}^*(f, s) - L_{\text{above}}^*(f, s) = 2\pi i e^{-\pi i s/2} \text{Res}_{\tau=\tau_p} [f(\tau) \tilde{k}_T(\tau, s)]. \quad (25)$$

We define the edge value as the principal value $L^* = \frac{1}{2}(L_{\text{above}}^* + L_{\text{below}}^*) = L_{\text{below}}^* - \pi i e^{-\pi i s/2} \text{Res}_{\tau=\tau_p} [f \tilde{k}_T]$. At the elliptic point $\tau_p = i$ the pole lies on the S -segment instead; its principal value, reached as $\tau_0 \rightarrow i$, is finite and basepoint-independent.

Proof. The two routings of γ^T differ by a contour encircling τ_p once, so the residue theorem gives $L_{\text{below}}^* - L_{\text{above}}^* = 2\pi i e^{-\pi i s/2} \text{Res}_{\tau=\tau_p} [f \tilde{k}_T]$, and the principal value is their mean. At $\tau_p = i$ the pole sits on the degenerate S -segment; displacing to $\tau_0 = i(1 + \delta)$ keeps it on γ^S for all small $\delta > 0$, so no wall is crossed and L^* is finite and δ -independent (verified numerically [4]; a closed proof is the S -segment principal value of [3]). $\square$

Corollary 3.10 (Wall-crossing). *By Proposition 3.1, $L^*(f, s)$ is fixed within each homotopy class. Crossing a wall, where the contour sweeps past an interior pole τ_p , changes it by*

$$\Delta L^*(f, s) = 2\pi i e^{-\pi i s/2} \text{Res}_{\tau=\tau_p} [f(\tau) \tilde{k}_T(\tau, s)] \quad (26)$$

when the T -segment crosses τ_p (and by the same with τ^{s-1} in place of $\tilde{k}_T$ when the S -segment does). Thus L^* is fixed by the contour up to the residues it crosses.

Proof. The within-class constancy is Proposition 3.1. Crossing a wall, the contour sweeps once past τ_p ; the residue theorem then adds $2\pi i e^{-\pi i s/2} \text{Res}_{\tau=\tau_p}[f \widetilde{k}_T]$ when the T -segment crosses it, or the same with τ^{s-1} when the S -segment does. $\square$

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